Gauge Structure and Architectural Inductive Bias for Axial Parity Codes: A Theorem and a Negative Result

Brian Guarraci
Galatheus
brian@galatheus.com

October 2026 revision

Abstract

We study the design of learned decoders for sparse-binary voxel compression with an axial parity code. We prove that the codespace of axial-line, plane, and global parity on the 3-torus is exactly the image of the discrete third partial derivative $\partial_x\partial_y\partial_z$ over $\mathbb{F}_2$, with parity-check rank $3n^2 - 3n + 1$. The potential parameterization has an exact redundancy given by the kernel of this derivative operator. This is an algebraic description of the classical codespace; an equivalence to a particular physical fracton gauge theory is not established. The proof is elementary once the lattice is read through its tensor-product structure, and the theorem caught a real indexing bug in our previously deployed parity matrix (reported rank 2016 instead of the correct 2977 at $n = 32$).

The natural architectural reading of this theorem — parameterise the network's residual as $r = \partial_x\partial_y\partial_z\phi$ and predict the gauge potential ϕ — fails decisively. Across 14 configurations spanning capacity, training time, and several optimisation tricks, this *phi-prior* decoder plateaus at validation IoU ≈ 0.29 versus ≈ 0.48 for an unconstrained baseline at matched configuration; the gap persists across the explored configurations, although matched multi-seed tests of all architectures remain to be performed. A soft U(1) latent-gauge variant does better (≈ 0.51), but removing its auxiliary gauge losses does not reduce validation performance in the reported single-seed ablation. A generic multi-scale autoencoder obtains the highest reported validation score (0.572 ± 0.004 over three seeds). The empirical conclusion is specific to these implementations and budgets. The principal lesson is methodological: a known mathematical structure characterises *what is reachable*, not the right inductive bias for a learned decoder, and enforcing exact algebraic structure architecturally can actively harm optimisation — a concrete coding-theoretic instance of the broader finding that hard equivariance/constraint enforcement often underperforms a soft prior. We report validation metrics on ShapeNet-32^3 and flag a held-out, multi-seed cross-domain study (including surface-code QEC) as future work.

1 Introduction

The motivating communication problem is to transmit a syndrome that restricts a receiver's learned source prior to a compatible set of reconstructions. A prior ranks alternatives; it does not make every underdetermined syndrome uniquely invertible. In the experiments here, the parity codec for sparse-binary voxel compression encodes a 32^3 binary occupancy field by transmitting (a) a syndrome consisting of parity sums along axis-aligned lines, planes, and the volume total, and (b) optional spectral side-information (low-frequency DCT coefficients of the voxel field). At the receiver, a learned network reconstructs the voxel field given the syndrome and side-information. This is a clean instance of *syndrome-conditioned reconstruction on a regular lattice* and shares its abstract structure with surface-code quantum error correction, where stabiliser measurements play the role of parity checks.

Within the companion paper family, this note is the algebraic and negative-control member. The compression paper asks which checks help a learned proposal; the surface-code and X-cube papers test geometry-aware neural decoders on quantum codes. Here we prove the algebraic structure behind the axial-parity codec and then test the tempting architectural conclusion one might draw from that proof. The main lesson is that a correct mathematical description of the constraint set is not, by itself, a recipe for a good learned decoder.

We pursue three connected goals:

1. **Characterise the codespace algebraically.** What is the structure of the set of voxel fields satisfying every parity check? We give a theorem (§3) stating that the codespace is exactly the image of the discrete third partial derivative over $\mathbb{F}_2$. This provides a constructive parameterisation with a gauge-like redundancy: every codeword is the discrete-XOR of eight corner values of an underlying scalar potential ϕ.
2. **Test whether the theorem prescribes a useful architecture.** The most direct interpretation is to predict ϕ with a neural network and emit $r = \partial_x\partial_y\partial_z\phi$ before reconstructing $x_{\text{base}} \oplus r$. We call this the *phi-prior* decoder. We find it fails decisively (§5) — across capacity, training time, and several optimisation tricks, it plateaus at ≈ 0.29 IoU versus an unconstrained baseline's ≈ 0.48 at matched compute.
3. **Find what does work.** A multi-scale latent autoencoder (§6) — encoder $\to 8^3$ latent $\to$ SharpNet upsample, with no gauge structure of any kind — beats both the phi-prior and the unconstrained baseline. The comparisons suggest that multi-scale processing is useful and that the tested gauge restrictions are unnecessary for these validation scores. Differences in parameter count and limited seed coverage prevent an exhaustive causal attribution.

The intended contribution is therefore a theorem, a negative result, and a controlled ablation, organised around the question of when a known mathematical structure should and should not be enforced architecturally in a learned decoder. A natural follow-up — whether the same architectural lesson transfers to other syndrome-inference tasks (in particular surface-code QEC) — is discussed as a deferred, multi-seed cross-domain study in §7.

2 Background

2.1 Axial parity codec

Let $\Lambda = (\mathbb{Z}/n)^3$ be a periodic 3-cubic lattice with voxel field $x : \Lambda \to \mathbb{F}_2$. The codec defines three families of parity checks on x:

- **Line.** $L^a_{\alpha\beta}(x) = \sum_t x_{t\hat{e}_a+\alpha\hat{e}_b+\beta\hat{e}_c} \pmod 2$ for each axis $a \in \{x, y, z\}$ and perpendicular coordinates (α, β).
- **Plane.** $P^a_\alpha(x)$ summing over the plane perpendicular to a at coord α.
- **Global.** $G(x) = \sum_v x_v$.

These checks are stacked into a parity matrix $H : \mathbb{F}_2^{n^3} \to \mathbb{F}_2^M$. For $n = 32$ the bug-fixed implementation has rank 2977. The codespace is $\mathcal{C}(n) = \ker H$.

The encoder transmits the syndrome Hx together with a low-frequency DCT field of x (variable bit budget; we use levels $k \in \{4, 5, 6, 7\}$ giving $\{2744, 5832, 10648, 17576\}$ bits respectively). The decoder constructs a canonical base solution x_{base} from the syndrome and a learned residual $r = x \oplus x_{\text{base}}$; the residual lies in the codespace by construction since x and x_{base} share the same syndrome. The learning task is to predict r from x_{base}, the syndrome, and (optionally) the DCT side-info.

2.2 Surface-code QEC

Surface and toric codes are $\mathbb{Z}_2$ stabilizer codes with a connection to lattice gauge theory [4, 5]. Their decoders infer logical effects from a syndrome under a noise prior. The companion

experiment uses a rotated planar surface code with repeated measurements represented on a spatiotemporal detector grid. This is statistically related to prior-conditioned reconstruction, but its target is a logical class rather than a voxel field. No physical equivalence between the axial codec and these quantum codes is assumed.

2.3 Notation

$(\partial_a \phi)_v := \phi_v + \phi_{v+\hat{e}_a} \pmod 2$ is the discrete forward difference over $\mathbb{F}_2$ in direction a.

3 Theorem: codespace structure of the axial parity code

Theorem 1. *For all $n \geq 1$ on the 3-torus $\Lambda = (\mathbb{Z}/n)^3$,*

$$\mathcal{C}(n) = \text{image}(\partial_x \partial_y \partial_z) \subseteq \mathbb{F}_2^{n^3}, \qquad \text{rank}_{\mathbb{F}_2}(H) = 3n^2 - 3n + 1.$$

Proof. Write $V = \mathbb{F}_2^n$ and identify $\mathbb{F}_2^{n^3} = V \otimes V \otimes V$, the three factors indexed by the x, y, z coordinates. The one-direction forward difference acts on a single tensor factor: $\partial_x = \partial \otimes I \otimes I$, $\partial_y = I \otimes \partial \otimes I$, $\partial_z = I \otimes I \otimes \partial$, where $\partial = I + S$ and S is the cyclic shift on V. Hence $T := \partial_x \partial_y \partial_z = \partial \otimes \partial \otimes \partial$.

Step 1 (codespace as an orthogonal complement). Each line, plane, and global check equals the $\mathbb{F}_2$ inner product of x with the indicator vector of an axis-parallel line, an axis-normal plane, or the whole lattice, respectively. Thus $\mathcal{C}(n) = \ker H = R^\perp$, where $R \subseteq \mathbb{F}_2^{n^3}$ is the span of all check indicators. A plane is the $\mathbb{F}_2$-sum of the parallel lines it contains, and the global vector is the sum of parallel planes, so R is already spanned by the axis-parallel line indicators alone.

Step 2 (the span of lines). The indicator of the x-line at transverse coordinates (β, γ) is $\mathbf{1} \otimes \delta_\beta \otimes \delta_\gamma$, where $\mathbf{1} \in V$ is the all-ones vector and δ_j are standard basis vectors. As (β, γ) range over $[n]^2$ the vectors $\delta_\beta \otimes \delta_\gamma$ span $V \otimes V$, so the x-lines span $\langle \mathbf{1} \rangle \otimes V \otimes V$. This is exactly $K_x := \{f : \partial_x f = 0\}$, the fields constant along x. Likewise the y- and z-lines span K_y and K_z, so $R = K_x + K_y + K_z$.

Step 3 (dimension). Choose any basis B of V with $\mathbf{1} \in B$. Then K_x, K_y, K_z are each spanned by a subset of the common tensor basis $B \otimes B \otimes B$, hence are coordinate subspaces, so inclusion–exclusion for dimensions is exact. With $\dim K_a = n^2$, $\dim(K_a \cap K_b) = n$ (fields constant in two axes), and $\dim(K_x \cap K_y \cap K_z) = 1$ (global constants),

$$\dim R = 3n^2 - 3n + 1.$$

Therefore $\text{rank}_{\mathbb{F}_2}(H) = \dim R = 3n^2 - 3n + 1$ and $\dim \mathcal{C}(n) = n^3 - (3n^2 - 3n + 1)$.

Step 4 (kernel of T). For a tensor product of operators acting on separate factors, $\ker(\partial \otimes \partial \otimes \partial) = \ker \partial \otimes V \otimes V + V \otimes \ker \partial \otimes V + V \otimes V \otimes \ker \partial$. On the n-cycle, $\ker \partial = \langle \mathbf{1} \rangle$ (a vector killed by $I + S$ is shift-invariant, hence constant). Thus $\ker T = K_x + K_y + K_z = R$, and $\dim \text{image}(T) = n^3 - \dim \ker T = n^3 - (3n^2 - 3n + 1) = \dim \mathcal{C}(n)$.

Step 5 (equality). For any ϕ, every axis-aligned line/plane/global sum of $T\phi = \partial_x \partial_y \partial_z \phi$ telescopes to 0 (mod 2) under the periodic boundary, since each ϕ_v enters such a sum an even number of times; hence $\text{image}(T) \subseteq \mathcal{C}(n)$. Two subspaces, one contained in the other and of equal finite dimension, coincide. Therefore $\mathcal{C}(n) = \text{image}(T)$. □

Empirical verification. We verified the theorem by brute-force $\mathbb{F}_2$ Gaussian elimination on the full check matrix at $n \in \{4, 6, 8\}$ (`verify_fracton_theorem.py`), obtaining $\text{rank}(H) \in \{37, 91, 169\}$ in exact agreement with $3n^2 - 3n + 1$, $\text{rank}(T) \in \{27, 125, 343\}$ in agreement with $n^3 - (3n^2 - 3n + 1)$, and $\dim \mathcal{C}(n) = \text{rank}(T)$ with random ϕ producing codewords that satisfy every check (50/50 per n), confirming $\mathcal{C}(n) = \text{image}(T)$ as a subspace. The $n = 32$ arithmetic gives $3 \cdot 1024 - 96 + 1 = 2977$.

The bug the theorem caught. The previously deployed parity matrix reported rank 2016 at $n = 32$ rather than 2977. Running its construction through the same verifier yields rank $\in \{28, 66, 120\}$ at $n \in \{4, 6, 8\}$ — strictly smaller than $\{37, 91, 169\}$, hence a *different, lower-rank* code (a duplicate-axis indexing error, not a re-presentation of the same checks). The closed-form rank is thus also a practical correctness check for the codec's bit-budget accounting.

Interpretation and the scope of the gauge language. Writing $x = T\phi$ gives the exact redundancy $\phi \mapsto \phi + \kappa$ for $\kappa \in \ker T = K_x + K_y + K_z$. Here "gauge" refers to this non-unique potential representation. The line checks span all the constraints; plane and global checks are dependent combinations. The theorem is about a binary linear subspace, and does not supply a quantum Hamiltonian or a phase classification.

Subsystem conservation laws and redundant potentials motivate a connection to fracton models [1, 2, 3], but the cited rank-2 scalar-charge theory has tensor fields with Gauss law $\partial_i \partial_j E^{ij} = \rho$. Identifying the present scalar triple-difference construction with that theory requires an additional mapping of fields, local operators, and constraints. We do not provide that mapping or claim equivalence to the X-cube phase. The constructive codespace theorem and the decoder experiments stand independently of it.

4 Experimental setup

All experiments use ShapeNet-32^3 [6] binary voxels with the bug-fixed parity check matrix giving syndrome of dimension $M = 2977$. Training target is the residual $r = x \oplus x_{\text{base}}$, which lies in the codespace. Loss is binary cross-entropy on r, optionally with positive-class weighting. The reported metric is intersection-over-union (IoU) on the reconstructed voxel field $\hat{x} = x_{\text{base}} \oplus \hat{r}$, evaluated on a held-back validation set (not a disjoint test split; see §9).

Unless stated otherwise: train subset 2,048 samples, validation 512 samples, batch size 8, AdamW with lr 5×10^{-4}, cosine LR schedule. We report best validation IoU achieved during training, and multi-seed mean ± stdev where indicated.

Architectures compared:

- **x-prior** (baseline): ResNet3D at varying width/depth, predicts r directly via flat-resolution 3D convolution at constant spatial resolution, syndrome conditioning via MLP.
- **phi-prior:** same backbone, output reinterpreted as ϕ; final operator is the bilinear soft-XOR of 8 corner ϕ's with output bias init at -5 to break XOR symmetry.
- **gauge transporter:** U(1) latent-lattice gauge with parallel-transport reconstruction from a seed voxel, plus auxiliary curvature/transport regularisers.
- **latent autoencoder:** encoder $\to 8^3$ latent $\to$ SharpNet decoder, no gauge structure.

5 Hard architectural enforcement plateaus far below baseline

Across 14 experiments varying capacity, training time, and optimisation tricks (output bias init, positive-class weighting, residual escape hatch, straight-through estimator, multi-scale capacity), the phi-prior decoder plateaus at val_iou ≈ 0.29 vs the unconstrained x-prior at 0.48 at matched parameter count and training budget.

The plateau is robust to capacity (5M = 12M), training duration (20 = 30 epochs), positive weighting (sweet spot at pos_weight= 19), and data scale (2K = 8K). The phi-prior occasionally exhibits a phase-transition jump from ~ 0.13 to ~ 0.21 around epoch 12 under the right hyperparameters but always saturates at ~ 0.29 soon after. **The tested potential-output parameterization performs poorly under these training configurations.**

One possible optimization explanation concerns the soft-XOR output. Its partial derivatives are products of factors with magnitude at most one, which can attenuate gradients. Each

variant	params	epochs	pos_weight	val_iou
x-prior	5.0M	20	19	**0.481**
x-prior	12.0M	20	19	0.489
x-prior	17.9M	20	19	0.474
phi-prior, tanh-product XOR	1.7M	3	1	0.002
phi-prior, bilinear + bias init -5	1.7M	5	1	0.021
phi-prior, +pos_weight= 19	5.0M	20	19	0.283
phi-prior, 12M	12.0M	20	19	0.276
phi-prior, 30 epochs	5.0M	30	19	**0.291**
phi-prior, 8K train data	5.0M	25	19	0.292
phi-prior, residual escape hatch	5.0M	5	1	0.001
phi-prior, STE hard-XOR	1.7M	10	1	0.001

Table 1: Phi-prior decoder vs x-prior baseline (best validation IoU, single-seed per cell). The phi-prior plateau at ≈ 0.29 IoU is robust across capacity, training duration, and optimisation; the unconstrained x-prior at matched compute reaches ≈ 0.48. The robustness of the gap across 14 cells, not any single comparison, is the evidence.

potential variable also affects eight neighboring residual positions, coupling their corrections. These features motivate a credit-assignment hypothesis; the reported experiments do not isolate them from other optimization effects.

6 What works instead: a generic multi-scale prior

If hard gauge enforcement is the wrong inductive bias, what is the right one? A generic multi-scale latent autoencoder — encoder downsampling $32^3 \to 8^3$ with C_{latent} channels, then a SharpNet decoder upsampling $8^3 \to 32^3$ with residual blocks at each scale, and *no gauge structure of any kind* — beats both the phi-prior and the unconstrained x-prior under the reported training budgets (Table 2).

architecture	gauge structure	params	val_iou
phi-prior (hard $\mathbb{F}_2$, every voxel)	hard	5.0M	0.283
x-prior (unconstrained ResNet3D)	none	5.0M	0.481
soft U(1) transporter	soft, coarse latent	4.4M	0.509
— auxiliary gauge losses removed	soft, coarse latent	4.4M	0.518
multi-scale autoencoder	**none**	16M	0.572 ± 0.004

Table 2: Architecture comparison (best validation IoU on 512 held-back ShapeNet samples). The autoencoder row is the mean $\pm$ stdev over seeds $\{42, 43, 44\}$; the other rows are single-seed (the phi-prior plateau is corroborated by the 14-run sweep in Table 1). The multi-scale autoencoder has the highest reported score, but parameter counts and seed coverage differ across rows.

Two exploratory comparisons inform the architectural interpretation. First, the soft U(1) transporter with all auxiliary curvature/transport/latent-reconstruction losses *removed* (val_iou 0.518) slightly outperforms the full version (0.509): this single-seed comparison shows no benefit from the auxiliary regularisers. Second, dropping the parallel-transport operator entirely — leaving a plain multi-scale autoencoder — improves things further (4.1M: 0.554; 16M: 0.572). These results favor the tested multi-scale architectures, but a fully matched multi-seed ablation is needed to isolate the responsible components.

A single-seed bottleneck-size sweep indicates an 8^3 latent is the best of the tested configurations: 4^3 gives 0.562 at 32M parameters and 16^3 gives 0.542 at 3M. Since capacities also

differ, these measurements do not isolate bottleneck size. This is consistent with the interpretation that the bottleneck implements an implicit low-frequency/smoothness prior, matched to the sparse, surface-concentrated structure of the occupancy field — precisely the property the gauge constraint is orthogonal to. We treat the bottleneck-sweep numbers as single-seed exploratory evidence, not multi-seed claims.

7 A deferred cross-domain question

Classical voxel reconstruction and quantum logical decoding both combine syndrome evidence with a prior, but their targets and input statistics differ. The classical target is a sparse occupancy field; the quantum input is a detector field whose event density depends on the noise model. Representing it as a full 3D array does not make its events dense.

Preliminary architectural observations motivate testing whether aggressive spatial bottlenecks discard useful detector-position information. They are not retained here as cross-domain results. A defensible comparison requires disjoint test sets, multiple seeds, matched resource budgets, and paired evaluation where applicable. The present voxel architecture comparisons themselves use validation scores and incomplete multi-seed coverage; they do not yet meet that standard either.

8 Discussion

A theorem characterises feasibility, not inductive bias. The gauge theorem tells us *what is reachable* (the codespace), not *how to reach it* (the architecture). Architectural enforcement of the theorem's constructive parameterisation hurt; the reported validation scores favor a generic multi-scale architecture. This does not rule out other ways of exploiting the exact structure. The compression codec's existing DCT side-information implements a related smoothness prior explicitly, and the multi-scale autoencoder appears to learn one implicitly via its 8^3 bottleneck.

Relation to soft-constraint priors. The phi-prior failure is a concrete coding-theoretic instance of a more general phenomenon: hard architectural enforcement of an exact algebraic or equivariance structure can underperform a soft prior or an unconstrained pathway. Finzi et al.'s residual pathway priors [8] make this point for equivariance constraints and propose converting hard constraints into soft penalties; our soft U(1) transporter is the analogous move in this setting, and the ablation shows even its soft gauge losses are not what carries the win. The lesson is not that the gauge structure is wrong — it is exactly correct as a characterisation of the codespace — but that exactness of the constraint and usefulness as an inductive bias are different properties.

Open questions. (i) Can the gauge structure be exploited as a *post-hoc* projection or a *soft penalty* (rather than as architectural enforcement) without the optimisation cost we observed? (ii) The theorem generalises to higher dimensions (the codespace of axial parity on the d-torus is the image of $\partial_1 \cdots \partial_d$, with rank obtainable by the same tensor argument); does the architectural lesson generalise too? (iii) Does the source-field vs detector-field distinction sketched in §7 hold up under a rigorous multi-seed cross-domain test, and where do quantum LDPC codes with sparser stabilisers fall?

9 Limitations

We are explicit about scope. (1) All reported IoU is *validation* IoU on a held-back ShapeNet-32^3 split, not a disjoint held-out test split; the multi-seed autoencoder result uses seeds $\{42, 43, 44\}$,

while the x-prior, phi-prior, and transporter rows and the bottleneck-size sweep are single-seed exploratory measurements. (2) The comparison in Table 2 is matched in training budget but not exactly in parameter count across all rows; parameter counts vary substantially, and the single-seed baselines do not establish their training variability. Equal epochs or samples also do not imply equal FLOPs. The results motivate, but do not replace, matched-capacity, matched-compute, multi-seed controls. (3) Results are at 32^3 resolution on a single object dataset. (4) The cross-domain QEC discussion (§7) is a hypothesis, not a measured result.

10 Conclusion

We proved a clean algebraic characterisation of the parity codec's codespace — it is exactly the image of the discrete third derivative $\partial_x\partial_y\partial_z$ over $\mathbb{F}_2$, with closed-form rank $3n^2-3n+1$ — and we used it to catch an indexing bug in a deployed parity matrix. We then tested the natural architectural translation of the theorem (predicting the gauge potential and applying the third derivative as a fixed output operator) and found it catastrophically worse than an unconstrained baseline. The actual win on the compression task comes from a generic multi-scale autoencoder with no gauge structure, and the exploratory ablations do not show a benefit from the tested auxiliary gauge losses. They do not establish a general failure of algebraically structured neural decoders.

The empirical conclusion is specific to these implementations and budgets. The principal lesson is methodological: a known mathematical structure characterises feasibility but does not prescribe the right inductive bias; architectural choice should be governed by the optimisation landscape and signal structure of the specific task. We provide a verified theorem, a clean negative example of constraint enforcement, and initial ablations that motivate further mechanism tests.

Reproducibility. All experiments are reproducible from the released code. In the family repository, the theorem verifier is grouped with the X-cube fracton code, and the gauge negative-result experiments are preserved as their own reproducible submodule. Run-by-run configuration and results, including discarded approaches, are catalogued in `experiments.tsv`. The theorem verifier (`verify_fracton_theorem.py`) reproduces the rank claims at $n \in \{4, 6, 8\}$ in ~ 20 s on CPU. See the companion `REPRODUCIBILITY.md` for exact commands.

References

[1] M. Pretko. Subdimensional particle structure of higher rank U(1) spin liquids. *Phys. Rev. B*, 95:115139, 2017.

[2] S. Vijay, J. Haah, and L. Fu. Fracton topological order, generalized lattice gauge theory, and duality. *Phys. Rev. B*, 94:235157, 2016.

[3] J. Haah. Local stabilizer codes in three dimensions without string logical operators. *Phys. Rev. A*, 83:042330, 2011.

[4] F. Wegner. Duality in generalized Ising models and phase transitions without local order parameters. *J. Math. Phys.*, 12:2259, 1971.

[5] A. Y. Kitaev. Fault-tolerant quantum computation by anyons. *Annals of Physics*, 303(1):2–30, 2003. arXiv:quant-ph/9707021 (1997).

[6] A. X. Chang et al. ShapeNet: An Information-Rich 3D Model Repository. arXiv:1512.03012, 2015.

[7] K. He, X. Zhang, S. Ren, and J. Sun. Deep residual learning for image recognition. In *CVPR*, 2016.

[8] M. Finzi, G. Benton, and A. G. Wilson. Residual pathway priors for soft equivariance constraints. In *NeurIPS*, 2021. arXiv:2112.01388.